Direction-Specific Traffic Congestion Hotspot Analysis on the O-4 / TEM Highway

Between Kurtköy and the Istanbul Finance Center Using TeslaMate Vehicle Telemetry

DSA210 Introduction to Data Science — Final Report

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Research question

How do speed patterns and recurring slow-traffic zones vary across time windows and travel directions on the selected O-4 / TEM corridor between Kurtköy and the Istanbul Finance Center, based on TeslaMate telemetry from a single vehicle?

170,684 analysis-window observations	45 km/h slow-speed threshold	7 DBSCAN low-speed zones	Mar 12–May 4, 2026 route-filtered period

1 Motivation

My daily commute to and from the university frequently uses the O-4 / TEM highway. Over time, I noticed that some parts of this route slow down more sharply than others, and that the pattern does not feel identical in both directions. This made the route a suitable data science problem because the question came from a repeated real-world observation rather than from a randomly chosen dataset.

The project uses my own vehicle telemetry to investigate sudden slowdowns and accelerations along the O-4 corridor between Kurtköy and the Istanbul Finance Center. TeslaMate makes this possible because it records timestamped GPS and speed observations during actual trips. Instead of relying on subjective trip impressions, the analysis tests whether repeated slowdowns appear as measurable spatial and temporal patterns.

The goal is not to describe all Istanbul traffic. The scope is narrower: use one vehicle's repeated corridor observations as a small traffic sensor and identify where and when slow movement appears after careful route filtering.

2 Research Question

> How do speed patterns and recurring slow-traffic zones vary across time windows and travel directions on the selected O-4 / TEM corridor between Kurtköy and the Istanbul Finance Center, based on TeslaMate telemetry from a single vehicle?

- Do mean speeds differ across Morning Peak, Midday, and Evening Peak?
- Does the proportion of slow observations differ across those time windows?
- Do low-speed observations form dense spatial clusters along the corridor?
- Are DBSCAN hotspot results stable under alternative slow-speed thresholds?
- What sampling, privacy, and causal-inference limits affect the interpretation?

3 Data Source and Collection

The data source is TeslaMate, an open-source, self-hosted vehicle telemetry system that records vehicle positions in a local PostgreSQL database. During active driving, the position stream is dense enough to capture detailed speed changes along the route.

The analysis uses timestamp, latitude, longitude, and speed from the TeslaMate positions table. The raw positions table contained 676,289 rows. After route filtering and time-window selection, the final analysis sample contains 170,684 observations from March 12, 2026 06:16:43 to May 4, 2026 07:08:32.

Timestamped GPS points can reveal personal routines and sensitive locations. For that reason, the analysis is presented through route-level filters, aggregate tables, and corridor-level figures rather than full raw trajectories.

4 Data Cleaning and Route Filtering

The first cleaning step keeps records with non-null timestamp, latitude, longitude, and speed values. Timestamps and numeric fields are parsed, speeds are restricted to the plausible 0–200 km/h range, and exact duplicate rows are removed. A broad O-4 / TEM candidate bounding box then removes non-corridor driving.

Route filtering is the central methodological step. I estimate the mainline corridor from night high-speed observations above 80 km/h, because those points are more likely to represent free-flowing highway travel rather than ramps, stops, or congestion. After transforming coordinates to UTM, DBSCAN identifies the largest high-speed mainline cluster. Median latitude/longitude waypoints are then created across longitude bins, and final route observations are kept if they fall within 50 m of this data-derived centerline.

Direction is inferred at the trip-like group level. Consecutive observations separated by more than 600 seconds start a new group. If movement along the route position increases by more than 1 km, the group is labeled Eastbound; if it decreases by more than 1 km, it is labeled Westbound; otherwise it is labeled Undetermined. Time windows are Morning Peak (06:00–10:30), Midday (10:30–14:30), and Evening Peak (15:00–20:00). Slow observations are defined as speed below 45 km/h.

Table 1. Filtering and analysis-stage row counts. The night high-speed row is a reference-building subset, not a linear retained stage.

Stage	Rows	Share of raw rows
Raw rows in positions table	676,289	100.0%
Rows loaded after non-null SQL filter	676,286	100.0%
Rows after timestamp and numeric parsing	676,286	100.0%
Rows after speed validation (0–200 km/h)	676,286	100.0%
Rows after duplicate removal	676,026	100.0%
Rows inside broad O-4/TEM candidate bounding box	300,961	44.5%
Night high-speed rows used to estimate route centerline	4,852	0.7%
Rows inside final 50 m mainline buffer	193,930	28.7%
Rows with inferred direction other than Undetermined	193,114	28.6%
Rows in morning/midday/evening analysis windows	170,684	25.2%
Slow rows in analysis windows (speed < 45 km/h)	62,376	9.2%

The final corridor sample is a filtered subset of the full telemetry. The broad bounding box removes non-corridor driving, and the 50 m buffer makes the final sample specific to the selected O-4 / TEM mainline. Within the final morning/midday/evening sample, 62,376 of 170,684 observations are below 45 km/h, so the slow-observation share is approximately 36.5%.

5 Exploratory Data Analysis

5.1 Route Speed Map

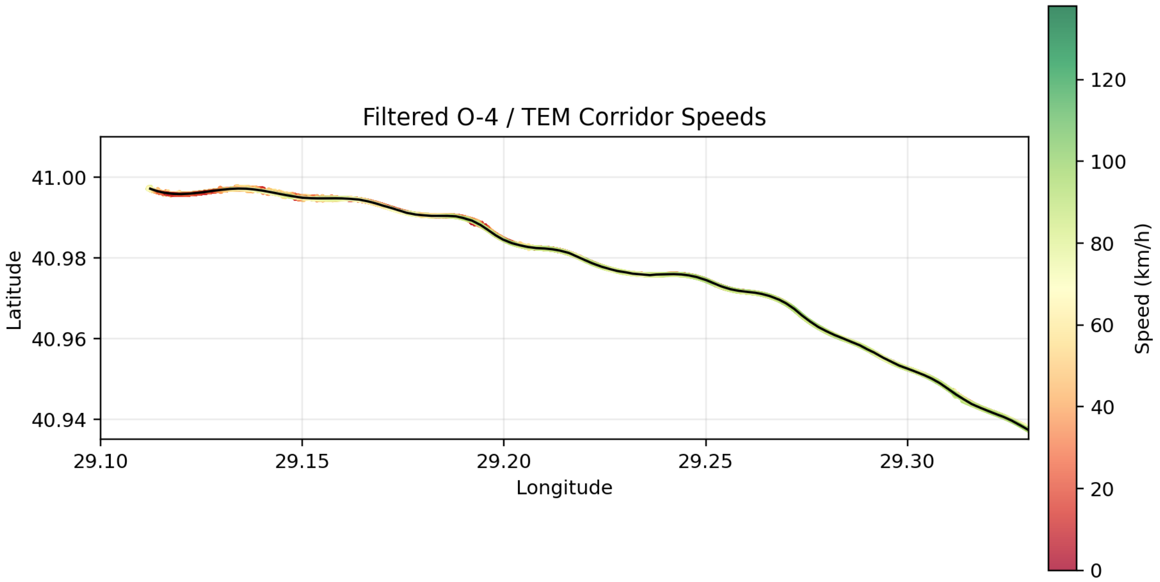

Figure 1. Filtered O-4 / TEM corridor observations colored by speed.

The route speed map shows that the filtered observations follow the selected O-4 / TEM corridor. High-speed and low-speed observations both appear along the route, which supports treating congestion as a spatial pattern rather than only as a single route-level average.

5.2 Time-Window Speed Patterns

Table 2. Speed and slow-share summary by time window.

Time window	Count	Mean	Median	Std. dev.	IQR	Slow share
Morning Peak	60,719	67.16	77.0	32.39	51.0	27.7%
Midday	57,006	55.31	56.0	30.66	55.0	40.2%
Evening Peak	52,959	53.79	53.0	31.16	56.0	42.8%

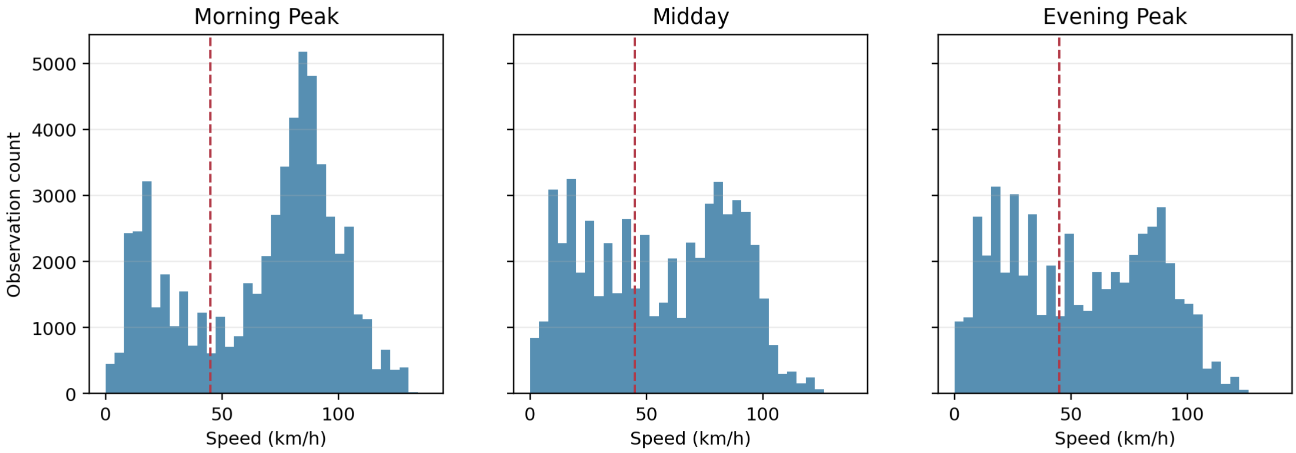

Figure 2. Speed distributions by time window with the 45 km/h slow-speed threshold marked.

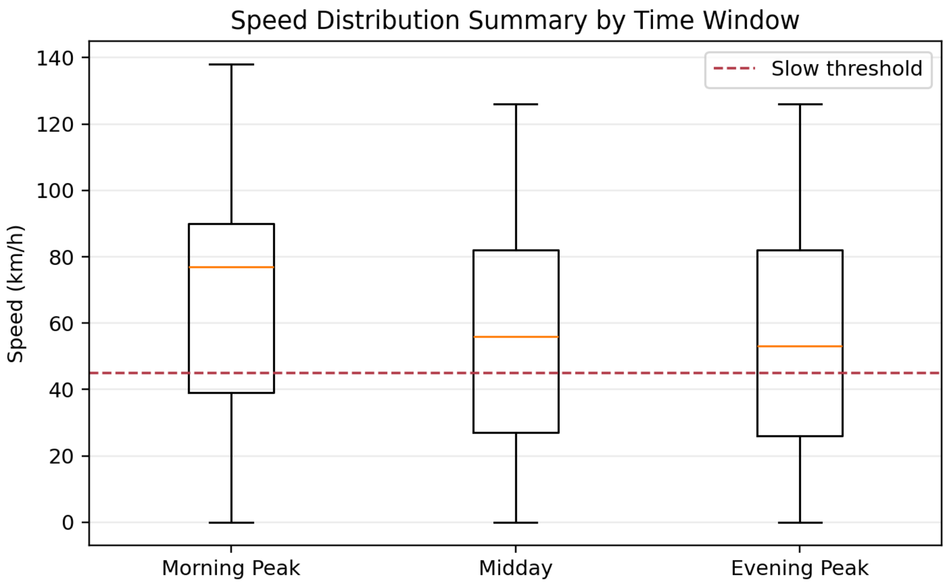

Figure 3. Boxplot comparison of speed by time window.

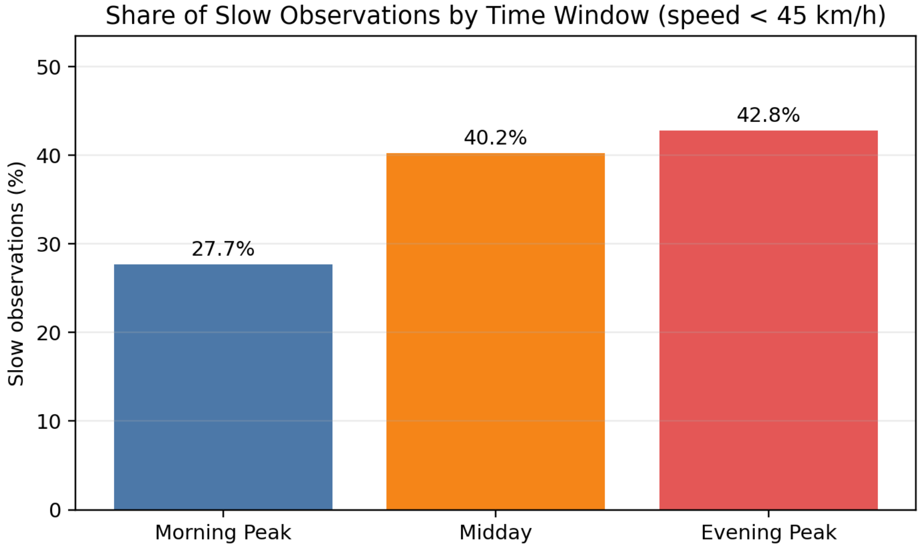

Figure 4. Share of observations below 45 km/h by time window.

Morning Peak has the highest mean and median speed in the final sample. Midday and Evening Peak have lower centers and higher slow-observation shares. The histograms show that the three windows have different speed distributions, the boxplot shows lower medians for Midday and Evening Peak, and the slow-share bar chart shows the clearest traffic interpretation: slow observations are more frequent in Midday and Evening Peak than in Morning Peak.

This does not mean Morning Peak is generally uncongested on the O-4 / TEM. It means that Morning Peak was faster in this vehicle's filtered observations. The result should be read together with the direction imbalance in the next section.

5.3 Direction-Specific Summary

Table 3. Direction-level speed summary after excluding Undetermined trips.

Direction	Count	Mean	Median	Std. dev.
Eastbound	90,167	65.78	76.0	32.18
Westbound	102,947	53.27	51.0	30.34

Table 4. Mean speed by direction and time window. Zero-count direction/window combinations are omitted.

Direction	Time window	Count	Mean	Median
Eastbound	Morning Peak	59,192	66.71	77.0
Eastbound	Midday	20,048	69.93	79.0
Undetermined	Midday	433	53.45	53.0
Undetermined	Evening Peak	383	47.57	51.0
Westbound	Morning Peak	1,527	84.37	85.0
Westbound	Midday	36,525	47.31	43.0
Westbound	Evening Peak	52,576	53.84	53.0

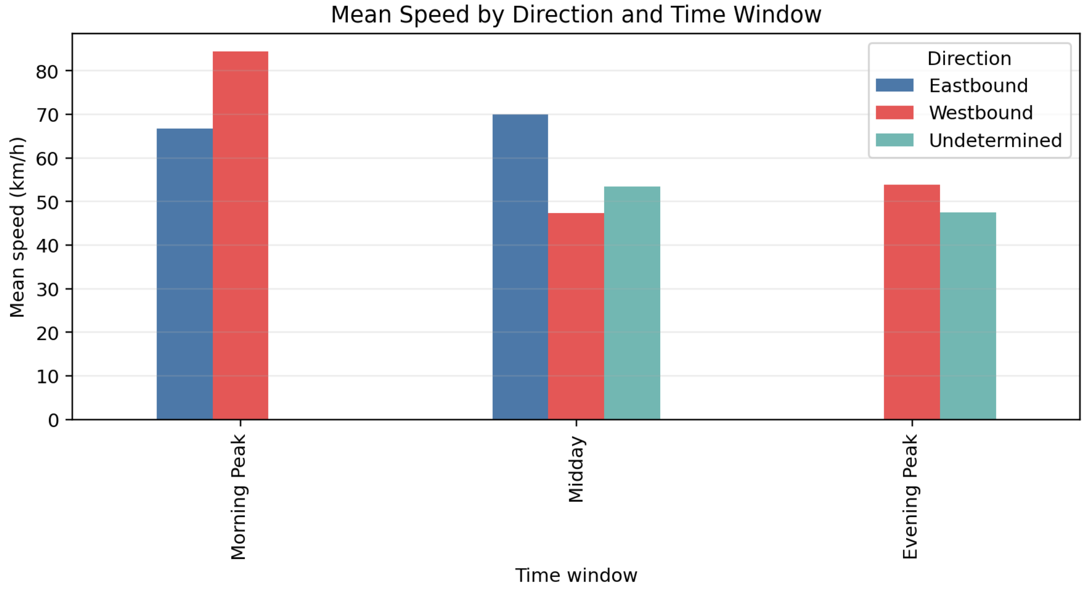

Figure 5. Mean speed by inferred direction and time window.

Direction changes the interpretation. Eastbound observations have a higher overall mean and median speed than Westbound observations. However, the direction-by-window table is uneven: Westbound dominates the Evening Peak sample, and there are no Eastbound Evening Peak rows in the final direction-time summary.

Table 4 includes 816 analysis-window observations whose route displacement was too small or ambiguous to assign a confident direction: 433 Midday rows and 383 Evening Peak rows. The table therefore reconciles to the full 170,684-row analysis-window sample, but direction should not be interpreted as a balanced two-way traffic comparison.

6 Formal Hypothesis Testing

6.1 Mean Speed Across Time Windows

Let μ_M, μ_D, and μ_E denote the true mean speeds for Morning Peak, Midday, and Evening Peak within the sampled corridor-driving process.

H_0: $\mu_M = \mu_D = \mu_E$.

H_A: At least one time-window mean speed differs.

The main test is a one-way ANOVA on observation-level speeds. Because consecutive telemetry points within the same drive are correlated, the p-values should be interpreted as evidence for differences in the observed telemetry process, not as independent-driver population inference. A Kruskal-Wallis test is added as a nonparametric robustness check.

Table 5. Formal hypothesis-test results.

Test	Statistic	Value	df	p-value	Decision
One-way ANOVA	F	3162.09	2, 170681	< 0.001	Reject H0
Kruskal-Wallis	H	6049.87	2	< 0.001	Reject H0
Chi-square independence	χ^2	3269.45	2	< 0.001	Reject H0

At $\alpha = 0.05$, the ANOVA rejects equal mean speeds across the three time windows. The estimated eta-squared is 0.0357, so time window explains a modest but meaningful share of observation-level speed variation. The Kruskal-Wallis test also rejects equal distributions, which supports the conclusion that the time-window pattern is not only an artifact of comparing means.

Table 6. Bonferroni-corrected pairwise Welch tests.

Comparison	t	Mean difference	Adjusted p-value	Decision
Morning Peak vs Midday	64.45	11.84	< 0.001	Reject H0
Morning Peak vs Evening Peak	70.82	13.36	< 0.001	Reject H0
Midday vs Evening Peak	8.15	1.52	< 0.001	Reject H0

All three time-window pairs differ statistically after correction. The largest practical differences are Morning Peak versus Midday and Morning Peak versus Evening Peak. Midday versus Evening Peak is statistically significant, but the difference in means is much smaller.

6.2 Slow-Observation Proportion Across Time Windows

Slow status is defined as speed below 45 km/h.

H_0: Time window and slow/not-slow status are independent.

H_A: Time window and slow/not-slow status are associated.

Table 7. Observed time-window by slow-status contingency table.

Time window	Not slow	Slow
Morning Peak	43,911	16,808
Midday	34,083	22,923
Evening Peak	30,314	22,645

The expected counts under independence are all far above 5; the smallest expected count is approximately 19,354. The chi-square expected-count condition is therefore satisfied. The chi-square test rejects independence, $\chi^2(2) = 3269.45$, $p < 0.001$. Cramer's V = 0.138, indicating a real but not overwhelming association between time window and slow status.

The traffic interpretation is that slow observations are not distributed uniformly across time windows. Morning Peak has the lowest slow share at 27.7%, while Midday and Evening Peak have slow shares of 40.2% and 42.8%.

6.3 Practical Significance

The large sample size makes very small p-values likely even when differences are moderate. The practical signal comes from the size and direction of the differences. A Morning Peak mean around 67 km/h compared with Midday and Evening means around 55 km/h and 54 km/h is meaningful in traffic terms. The slow-share gap is also meaningful: Midday and Evening Peak contain roughly 12–15 percentage points more slow observations than Morning Peak.

7 DBSCAN Hotspot Analysis

DBSCAN is used because the objective is to find dense spatial groupings of repeated low-speed observations, not to predict a label. The model can identify clusters with arbitrary shapes and label isolated slow points as noise.

The DBSCAN input is the set of analysis-window observations with speed below 45 km/h. Coordinates are converted to radians and clustered with haversine distance using Earth radius 6,371,000 m, eps = 120 m, and min_samples = 40. The model finds seven clusters and nine noise points.

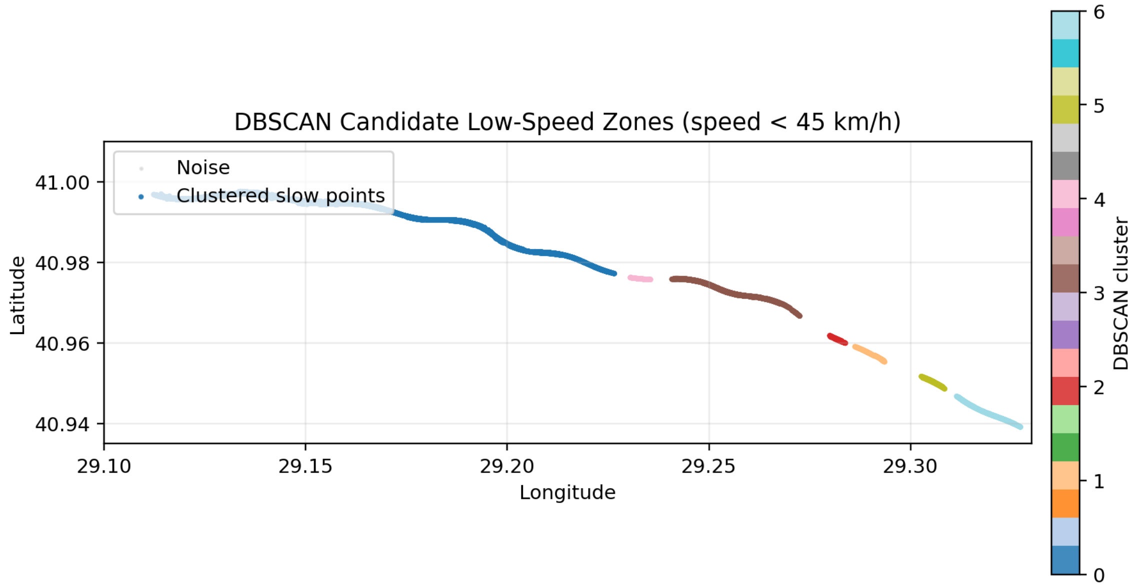

Figure 6. DBSCAN clusters of low-speed corridor observations.

Table 8. DBSCAN cluster summary. Mean route position and route span are measured along the data-derived corridor line.

Cluster	Points	Mean speed	Median speed	Mean route km	Span km
0	54,928	21.70	19.0	4.08	10.11
3	3,833	28.57	29.0	12.64	2.93
6	2,285	24.36	24.0	18.99	1.59
5	473	34.87	35.0	17.61	0.60
1	419	24.08	23.0	16.11	0.75
2	254	38.60	40.0	15.29	0.39
4	175	31.13	35.0	10.68	0.42

Table 9. Dominant time window and direction for each DBSCAN cluster.

Cluster	Dominant window	Dominant direction
0	Midday	Westbound
3	Midday	Westbound
6	Evening Peak	Westbound
5	Evening Peak	Westbound
1	Morning Peak	Eastbound
2	Morning Peak	Westbound
4	Morning Peak	Eastbound

The largest cluster contains most clustered slow points and has a median speed of 19 km/h. It spans about 10.11 km of route position, so it represents a long continuous low-speed region in the sampled corridor rather than a small isolated point. Several smaller clusters occur later along the route. The dominant windows and directions show that low-speed clustering is not evenly distributed across the sampled travel patterns.

These clusters are candidate recurring low-speed zones. They may correspond to merges, grades, ramps, recurring congestion, or sampling patterns. External incident, roadwork, weather, road-geometry, and traffic-flow data are not included, so the model identifies where slow points concentrate but does not prove why they occur.

8 Sensitivity Analysis

Table 10. DBSCAN sensitivity to the slow-speed threshold.

Threshold	Slow points	Clusters	Noise points	Largest cluster
40 km/h	56,570	7	0	50,251
45 km/h	62,376	7	9	54,928
50 km/h	68,063	7	37	59,314

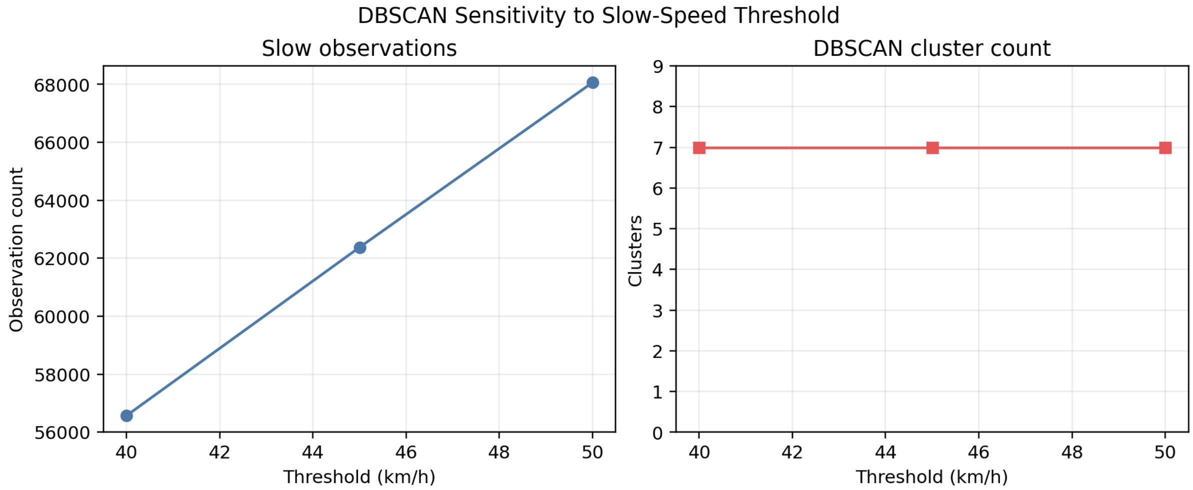

Figure 7. Slow-point and cluster-count sensitivity under 40, 45, and 50 km/h thresholds.

The cluster count remains stable at seven across all three slow-speed thresholds. The number of slow points increases as the threshold is relaxed from 40 to 50 km/h, which is expected. Noise points remain small compared with the total number of slow points. This supports the interpretation that the main low-speed spatial pattern is not created only by the exact 45 km/h threshold.

9 Findings

1. Speed patterns differ by time window in the filtered O-4 / TEM observations. Morning Peak has the highest mean and median speed in the final analysis run.

2. Slow-observation frequency is not uniform across time windows. Midday and Evening Peak have substantially larger slow-observation shares than Morning Peak.

3. Formal tests support the descriptive results: ANOVA, Kruskal-Wallis, and chi-square tests all reject their null hypotheses at $\alpha = 0.05$.

4. Direction matters, but direction-time coverage is uneven. Direction is useful for interpretation but not sufficient for a fully balanced two-way comparison.

5. DBSCAN identifies seven candidate recurring low-speed zones under the 45 km/h threshold.

6. Threshold sensitivity at 40, 45, and 50 km/h keeps the number of DBSCAN clusters stable, suggesting that the broad spatial pattern is robust to nearby slow-speed cutoffs.

The strongest conclusion is that this vehicle's filtered O-4 / TEM observations show meaningful time-window differences and repeated spatial concentration of low-speed observations. The analysis does not prove the cause of congestion, but it identifies where and when slow movement appears in the collected telemetry.

10 Limitations and Future Work

The dataset comes from a single vehicle and a non-random set of trips. The observations are repeated GPS measurements, not independent sampled drivers. Direction is inferred from movement along a data-derived route position and can be uncertain for short or ambiguous trips. GPS noise and the 50 m route buffer can affect which points enter the final route sample. The slow-speed threshold and

DBSCAN parameters affect cluster membership.

The analysis does not include external traffic, incident, construction, road geometry, weather, or public traffic-volume data. Therefore, it can identify observed speed patterns and candidate recurring low-speed zones, but it cannot make causal claims about why congestion occurs.

Future work should extend the data collection period, aggregate observations by trip or route segment before testing, add external traffic or weather data if available, improve map matching with road-network geometry, and compare results across more vehicles or public traffic datasets. A later version could also build segment-level time-series summaries or forecasting models after the observational and privacy constraints are handled carefully.

11 References

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